Sepehr Rezaee

sepehrrezaee2002@gmail.com | github.com/SepehrRezaee | My Scholar | linkedin.com/in/sepehr-rezaee/
sepehrrezaee.github.io

Education

Shahid Beheshti University, BS in Computer Sciences *2021 – 2025*

- **Interests:** Deep Learning, Computer Vision, Generative AI, and AI Safety

Allameh Tabatabaei (Advanced) High School, Math Diploma *2019 – 2021*

- **GPA:** 3.87/4.0

Experience

AI Specializing In Applied LLMs PropTy Global *2024 – Present*

- *Led the development of an AI-powered chatbot and home finder system using open-source LLMs and Retrieval-Augmented Generation (RAG) to enhance real estate customer service.*
- *Engineered a scalable backend infrastructure with FastAPI, PostgreSQL, and MongoDB, ensuring high reliability and performance for real-time interactions.*
- *Integrated and fine-tuned LLMs (GPT-Neo, GPT-J, LLaMA) for domain-specific tasks, significantly improving the accuracy and contextual relevance of chatbot responses.*
- *Successfully deployed the solution with Docker and Kubernetes, achieving seamless scalability and handling a large volume of concurrent user requests.*
- *Implemented real-time monitoring and continuous improvement strategies using Prometheus, Grafana, and the ELK stack, ensuring system stability and performance.*

Research Assistant, Robust and Interpretable Machine Learning Lab – Sharif University of Technology, Tehran *2024 – 2025*

- *Authored and co-authored 3 papers submitted to NeurIPS 2024, focusing on enhancing model reliability and security in machine learning.*
- *Developed and implemented 3 robust machine learning pipelines, increasing model reliability under adversarial conditions.*
- *Collaborated with a multidisciplinary team of 10 members to integrate machine learning solutions into real-world applications (Autonomous Driving, Face Detection, Diagnosing Disease), improving operational efficiency.*
- *Presented research findings at 2 international conferences, elevating the lab's visibility and fostering academic collaborations.*

Research Assistant, Artificial Intelligence and Scientific Computing Lab – Shahid Beheshti University, Tehran *2023 – 2025*

- *Co-authored 2 under-review & 1 published research papers, including:*
  - *Physics-Informed Lane-Emden Solvers Using Lynx-Net: Implementing Radial Basis Functions in Kolmogorov Representation*
  - *Leveraging Physics-Informed Convolutional Neural Networks (PICNNs) to Solve Linear and Non-linear Fokker-Planck Equations (FPEs)*
  - *Comparison of Pre-training and Classification Models for Early Detection of Alzheimer's Disease Using Magnetic Resonance Imaging*
- *Modeled disease progression using differential equations, enhancing the understanding of biological mechanisms.*
- *Employed Physics-Informed Neural Networks (PINNs), increasing model accuracy through the integration of physical laws.*

Deep Learning and Neuroscience Intern Researcher, Institute for Research in Fundamental Sciences (IPM) – Tehran *2023 – 2024*

- *Conducted comprehensive M/EEG data analysis utilizing advanced deep learning techniques to decode neural signals.*
- *Developed and optimized neural network architectures for improved signal processing and feature extraction.*
- *Collaborated with neuroscientists to interpret data results and contribute to the understanding of brain functionalities.*
- *Assisted in the preparation of research manuscripts and presentations for academic dissemination.*

Publications

- ***Scanning Trojaned Models Using Out-of-Distribution Samples*** (Accepted to NeurIPS) 2024

  Authors: Hossein Mirzaei, Ali Ansari, Bahar Dibaei Nia*, Mojtaba Nafez†, Moein Madadi†, **Sepehr Rezaee**†, Zeinab Sadat Taghavi, Arad Maleki, Kian Shamsaie, Mahdi Hajialilue, Jafar Habibi, Mohammad Sabokrou, Mohammad Hossein Rohban*

- ***A Contrastive Teacher-Student Framework for Novelty Detection Under Style Shifts*** (Submitted to ICLR) 2025

  *Authors: Hossein Mirzaei, Mojtaba Nafez, Moein Madadi, Arad Maleki, Mahdi Hajialilue, Zeinab Sadat Taghavi, **Sepehr Rezaee**, Ali Ansari,* *Bahar Dibaei Nia, Kian Shamsaie, Mohammadreza Salehi, Jafar Habibi, Mackenzie W. Mathis, Mahdieh Soleymani Baghshah, Mohammad Sabokrou, Mohammad Hossein Rohban*

- ***Comparison of Pre-Training and Classification Models for Early Detection of Alzheimer's Disease Using Magnetic Resonance Imaging*** (Accepted in ICCCCC 2023) — 2023

  *Authors: AH Karami, **S Rezaee**, E Mirzabeigi, K Parand*

- ***Hierarchical Clustering Algorithms, Chapter of Unsupervised Algorithms: Clustering (with Implementation)*** Aarvan Publications — 2022

  *Authors: Kourosh Parand, **Sepehr Rezaee**, et al.*

Selected Projects

AI Model Security: Enhancing Robustness Against Backdoors and Trojans — *2024*

- Developed methods to detect and mitigate backdoors in machine learning models, enhancing AI deployment security.
- Engineered algorithms using statistical analysis and pattern recognition, improving trojan detection rates.
- Contributed to NeurIPS 2024 publications, advancing the field of AI model security.
- **Tools Used:** Python, PyTorch, Scikit-learn, LaTeX

AI-Based Application for Early Detection of Alzheimer's Disease — *2023 – 2024*

- Designed and implemented a customized multi-modal model integrating biomedical and MRI datasets.
- Enhanced diagnostic accuracy through advanced machine learning techniques with Vision Language Models (VLMs).
- **Tools Used:** PyTorch, Hugging Face, OpenCV

Physics-Informed Neural Networks for Disease Progression Modeling — *2023*

- Created a Physics-Informed Neural Network integrating differential equations to predict disease progression accurately.
- Utilized clinical datasets and validated models with patient data, achieving higher accuracy than traditional methods.
- Published findings in peer-reviewed journals, contributing to AI-based healthcare innovations.
- **Tools Used:** PyTorch, NumPy, SciPy, Pandas

AI-Driven M/EEG Data Analysis for Neuroscience Research — *2022*

- Applied deep learning techniques to decode M/EEG signals, uncovering neural mechanisms.
- Streamlined data workflows by automating preprocessing and artifact removal, enhancing analysis efficiency.
- Facilitated insights into brain connectivity, supporting high-impact neuroscience research publications.
- **Tools Used:** MNE-Python, PyTorch, NumPy, Pandas

Teaching Assistant

Advanced Programming Head Teaching Assistant, Shahid Beheshti University, Tehran — 2024 – Present

Data Mining and Analysis Head Teaching Assistant, Shahid Beheshti University, Tehran — 2023

Basic Programming Teaching Assistant, Shahid Beheshti University, Tehran — 2022

Assistant Teacher and Mentor — 2022 – 2023

- Applications of Data Science and Artificial Intelligence in the Petrochemical Industry, the Water Industry & the Electricity Industry

Selected Courses

Courses: Foundations of Data Science (A^+, 1st), Data Mining (A^+, 1st), Advanced Data Mining (A^+, 1st), Foundation of Numerical Analysis (A^+, 1st), Non-Linear Optimization (A^+, 1st), Partial Differential Equations (A^+, 1st), Electromagnetics (A^+, 1st), Neural Network (A^+, 3rd), Foundation of Logic and Set Theory (A^+, 3rd), Principles of Operating Systems (A^+, 2nd), Foundations of Machine Learning (A^+, 2nd), Elements of Probability (A, 4th), Data Structures & Algorithms (A, 5th)

Skills

Programming Languages: Python, C++, C, MATLAB, C# & Java

Python Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, MNE-Python, NumPy, SciPy, Matplotlib, Scikit-Learn, NiPype, FastAPI, Django, Django REST Framework, Selenium

Other Tools and Technologies: JAX, PostgreSQL, NoSQL, MongoDB, Kotlin, , Git, Docker, Linux, Bootstrap

Interpersonal Skills: Problem Solving, Team Working

Languages: Fluent in Persian (speaking, reading, and writing), English (Professional working proficiency)

Reference Contacts

Prof. Kourosh Parand - k_parand@sbu.ac.ir
Prof. Mohammad Hossein Rohban - rohban@sharif.edu
Prof. Mohammad Sabokrou - mohammad.sabokrou@oist.jp